

automorphism representable by reals

Nathanael Chwojko-Srawley

December 20, 2025

Contents

1	Torsion-Free Automorphism Groups	7
1.1	The Main Theorem	7
1.2	The Abelian Case	8
1.2.1	The Inversion Trap	8
1.2.2	The Vector Space Barrier	8
1.3	The Non-Abelian Case	8
1.3.1	Central Automorphisms and Cohomology	9
1.3.2	The Scaling Contradiction	9
1.4	Conclusion	10
2	lie algebra stuff	13
2.1	Step 2: The Lie Algebra Connection	13
2.1.1	Step 3: Constructing the Scaling Automorphism	14
2.1.2	Step 4: The Contradiction	15
3	another idea	17
3.1	Introduction	17
3.2	Proof	17
4	another attempt	21
4.1	Theorem	21
4.2	Proof	21

A group G is [group] automorphism-representable if there exists a group H such that

$$\text{Aut}_{\mathbf{Grp}}(H) \cong G$$

From now on we'll just write Aut instead of $\text{Aut}_{\mathbf{Grp}}$.

Lemma 0.0.1: Rational Are Not Automorphism-representable

The rational numbers under addition $(\mathbb{Q}, +)$ are not automorphism representable

Proof:

The key to remember is that finitely generated subgroups of $\mathbb{Q}$ are cyclic. For the sake of contradiction say they are so that there exists a G such that:

$$\text{Aut}(G) \cong \mathbb{Q}$$

Then as:

$$\text{Inn}(G) \trianglelefteq \text{Aut}(G) \cong \mathbb{Q}$$

either $\text{Inn}(G)$ is a divisible group or cyclic. If it is is cyclic then it must have infinite order as $\mathbb{Q}$ is torsion free. Then as:

$$G/Z(G) \cong \mathbb{Z}$$

Elementary group theory tells us G is abelian. But if G is abelian there must be a torsion element $x \mapsto x^{-1}$ of order 2, a contradiction. On the other hand, if $G/Z(G)$ is not finitely generated, we can “cheat” by choosing any $gZ(G), hZ(G) \in G/Z(G)$ and generating the group H which is cyclic, so

$$\langle xZ(G) \rangle = \langle gZ(G), hZ(G) \rangle$$

Then it is easy to see that:

$$gh = x^n z_1 x^m z_2 = z^{n+m} z_1 z_2 = z^m z_2 x^n z_1 = hg$$

and as g, h were arbitrary, G is abelian giving us the same contradiction.

Another important fact: let us make sure that if G is an $\mathbb{F}_2$ -vector-space then $\text{Aut}(G)$ cannot be abelian. The point of this is that we want to abuse the $x \mapsto x^{-1}$ argument, but we can't do that if $x^{-1} = x$ and hence this is the identity:

Lemma 0.0.2: Vector Space, then Non-abelian

Assume $x \mapsto x^{-1}$ is the identity. Then G has an $\mathbb{F}_2$ -action, hence is a vector space and hence $\text{Aut}(G)$ cannot be abelian

Proof:

Say that $x \mapsto x^{-1}$ is the identity. Then every element has a natural $\mathbb{F}_2$ -action on it. Hence G is a vector-space which we know by linear algebra $\text{Aut}(G)$ is not abelian.

Hence, it truly is enough to check for $x \mapsto x^{-1}$. Let's label this σ .

Central automorphism and cohomology

Definition 0.0.3: Central Automorphisms

An automorphism $\alpha \in \text{Aut}(G)$ is called *central* if it induces the identity map on the central quotient $G/Z(G)$. Equivalently, for all $x \in G$, $x^{-1}\alpha(x) \in Z(G)$. In other words:

$$\alpha(x) \in xZ(G)$$

so α keeps elements in the cosets.

If we are trying to show $\text{Aut}(G) \not\cong \mathbb{Q}^n$ or $\mathbb{R}$, note that every automorphism *has* to be central, that is:

$$\text{Aut}_c(G) \cong \text{Aut}(G)$$

To show this, since $\text{Aut}(G)$ is abelian, the subgroup of inner automorphisms, $\text{Inn}(G) \cong G/Z(G)$, must be a normal subgroup of $\text{Aut}(G)$. Let $\alpha \in \text{Aut}(G)$ and $c_g \in \text{Inn}(G)$ (where $c_g(x) = gxg^{-1}$). Since $\text{Aut}(G)$ is abelian, α and c_g commute:

$$\alpha \circ c_g = c_g \circ \alpha$$

Applying this to all $x \in G$:

$$\begin{aligned} \alpha(gxg^{-1}) &= g\alpha(x)g^{-1} \\ \alpha(g)\alpha(x)\alpha(g)^{-1} &= g\alpha(x)g^{-1} \end{aligned}$$

Since α is surjective, as x varies over G , $\alpha(x)$ varies over all of G . Letting $y = \alpha(x)$, re-arranging we see that:

$$g^{-1}\alpha(g)y = yg^{-1}\alpha(g)$$

Letting $z = g^{-1}\alpha(g)$ we get:

$$zy = yz$$

Hence, as y varies through all elements of G ,

$$z = g^{-1}\alpha(g) \in Z(G) \quad \text{i.e.} \quad \alpha(g) \in gZ(G)$$

which is exactly the condition of a central automorphism. Thus if $\text{Aut}(G)$ is abelian, then

$$\text{Aut}(G) = \text{Aut}_c(G)$$

Now, each $\alpha \in \text{Aut}_c(G)$ gives rise to a derivation. In particular, the map $\tau_\alpha : G \rightarrow Z(G)$ defined by

$$\tau_\alpha(x) = x^{-1}\alpha(x)$$

is a derivation (i.e. 1-cocycle) which is easy to check as $Z(G)$ is abelian and hence these are even homomorphisms (see [1] for more details). Since $Z(G)$ is abelian and the action is trivial, this derivation is simply a homomorphism from G to $Z(G)$. Since the target is abelian, it factors through the commutator subgroup G' . Thus, we have an embedding:

$$\text{Aut}_c(G) \hookrightarrow \text{Hom}(G/G', Z(G))$$

The condition $\text{Aut}(G) = \text{Aut}_c(G)$ implies that *every* automorphism of G acts trivially on the quotient $Q = G/Z(G)$. This gives an interesting short exact sequence:

$$0 \rightarrow \text{Hom}(G/G', Z(G)) \rightarrow \text{Aut}_c(G) \rightarrow \text{Inn}(G) \cap Z(\text{Aut}(G))$$

Now, let $Q = G/Z(G)$. This notation is used as when we were working with $\text{Aut}(G) = \mathbb{Q}$ we had that Q was locally cyclic. In our case, this need not be the case, but it is still divisible and hence Q is suggestive of the useful qualities it possesses. As we are assuming G is non-abelian, $G/Z(G)$ is non-trivial and:

$$\omega : Q \times Q \rightarrow Z(G) \quad (a, b) \mapsto [a, b]$$

is non-trivial (if it were trivial, then for any choice of representative $xZ(G), yZ(G)$ of cosets we would get that $[x, y] = e$ which would give that G is abelian).

Torsion-Free Automorphism Groups

In this chapter, we explore a fascinating rigidity theorem concerning the structure of groups with "large" but abelian automorphism groups. Specifically, we investigate whether the automorphism group of a group G can be isomorphic to a torsion-free abelian group like the additive group of rational numbers $\mathbb{Q}$, the real numbers $\mathbb{R}$, or products thereof. The result is a striking non-existence theorem: no such group G exists. This proof weaves together basic group theory with concepts from homological algebra and the theory of nilpotent groups, highlighting the tension between the commutativity of $\text{Aut}(G)$ and the non-commutative structure required of G .

1.1 The Main Theorem

The central question is simple to state but requires careful analysis to answer: Does there exist a group G such that $\text{Aut}(G) \cong A$, where A is a torsion-free abelian group? The answer is negative.

Theorem 1.1.1: Non-Existence of Groups with Torsion-Free Abelian Automorphisms

Let A be a non-trivial torsion-free abelian group (e.g., $A = \mathbb{Q}$, $A = \mathbb{R}$, $A = \mathbb{Q} \times \mathbb{Q}$). There does not exist a group G such that $\text{Aut}(G) \cong A$.

The proof naturally divides into two cases based on the commutativity of G : the abelian case and the non-abelian case. Each case leads to a contradiction arising from the specific structural properties of A .

1.2 The Abelian Case

Let us first assume that G is abelian. In this setting, we can leverage the existence of a specific, well-behaved automorphism: the inversion map.

1.2.1 The Inversion Trap

Consider the map $\varphi : G \rightarrow G$ defined by $\varphi(x) = x^{-1}$. Since G is abelian, this map is a homomorphism:

$$\varphi(xy) = (xy)^{-1} = y^{-1}x^{-1} = x^{-1}y^{-1} = \varphi(x)\varphi(y)$$

Since every element in a group has a unique inverse, φ is a bijection, and thus $\varphi \in \text{Aut}(G)$. Observe that $\varphi^2(x) = (x^{-1})^{-1} = x$, so $\varphi^2 = \text{id}_G$. This means φ is an element of order 2 in $\text{Aut}(G)$ (unless $\varphi = \text{id}_G$, in which case its order is 1).

However, we assumed $\text{Aut}(G) \cong A$, and A is *torsion-free*. A torsion-free group contains no non-identity elements of finite order. Therefore, we must have $\varphi = \text{id}_G$. This implies:

$$x^{-1} = x \iff x^2 = e \quad \forall x \in G$$

A group satisfying $x^2 = e$ for all elements is known as an *elementary abelian 2-group*. Such a group carries the structure of a vector space over the field $\mathbb{F}_2$.

1.2.2 The Vector Space Barrier

If G is an elementary abelian 2-group, it is isomorphic to a direct sum of copies of $\mathbb{Z}_2$. Let V denote this vector space. The automorphisms of G are precisely the invertible linear transformations of V . Thus:

$$\text{Aut}(G) \cong \text{GL}(V)$$

We now analyze the structure of the General Linear Group $\text{GL}(V)$:

1. If $\dim(V) = 0$ or 1 (i.e., $|G| = 1$ or 2), then $\text{Aut}(G)$ is the trivial group. This contradicts the assumption that A is non-trivial (since A is isomorphic to $\mathbb{Q}$, $\mathbb{R}$, etc.).
2. If $\dim(V) \geq 2$, then $\text{GL}(V)$ is *non-abelian*. It is a standard result of linear algebra that for dimension ≥ 2 , there exist non-commuting matrices (linear transformations).

Since we assumed $\text{Aut}(G) \cong A$ is abelian, we have reached a contradiction. Thus, G cannot be abelian.

1.3 The Non-Abelian Case

We now turn to the more subtle case where G is non-abelian. Here, the contradiction arises from the interplay between central automorphisms and the existence of "scaling" automorphisms in torsion-free nilpotent groups.

1.3.1 Central Automorphisms and Cohomology

Assume G is non-abelian. Since $\text{Aut}(G)$ is abelian, the subgroup of inner automorphisms, $\text{Inn}(G) \cong G/Z(G)$, must be a normal subgroup of $\text{Aut}(G)$. Let $\alpha \in \text{Aut}(G)$ and $c_g \in \text{Inn}(G)$ (where $c_g(x) = gxg^{-1}$). Since $\text{Aut}(G)$ is abelian, α and c_g commute:

$$\alpha \circ c_g = c_g \circ \alpha$$

Applying this to an arbitrary $x \in G$:

$$\begin{aligned} \alpha(gxg^{-1}) &= g\alpha(x)g^{-1} \\ \alpha(g)\alpha(x)\alpha(g)^{-1} &= g\alpha(x)g^{-1} \end{aligned}$$

Since α is surjective, as x varies over G , $\alpha(x)$ varies over all of G . Thus, for the conjugation actions to be identical, the conjugating elements must differ by an element of the center:

$$\alpha(g)g^{-1} \in Z(G) \quad \forall g \in G$$

This condition defines a specific class of automorphisms known as *central automorphisms*, denoted $\text{Aut}_c(G)$.

Definition 1.3.1: Central Automorphisms

An automorphism $\alpha \in \text{Aut}(G)$ is called *central* if it induces the identity map on the central quotient $G/Z(G)$. Equivalently, for all $x \in G$, $x^{-1}\alpha(x) \in Z(G)$.

We have thus proven that if $\text{Aut}(G)$ is abelian, then $\text{Aut}(G) = \text{Aut}_c(G)$.

1.3.2 The Scaling Contradiction

We now demonstrate that for the groups in question, there must exist automorphisms that are *not* central, contradicting the previous section.

Since $\text{Inn}(G) \leq \text{Aut}(G)$ and $\text{Aut}(G)$ is torsion-free abelian, the quotient $Q = G/Z(G)$ is a torsion-free abelian group. Because G is non-abelian, the rank of Q must be at least 2 (if it were rank 1, i.e., locally cyclic, G would be abelian). We also have a non-trivial commutator map $\omega : Q \times Q \rightarrow Z(G)$ defined by $\omega(\bar{x}, \bar{y}) = [x, y]$.

In the context of torsion-free nilpotent groups (specifically class 2, which G must be), the structure is flexible enough to admit "scaling" automorphisms. For groups resembling the Heisenberg group over a field or ring that allows division (like subrings of $\mathbb{R}$), we can define automorphisms that act non-trivially on Q .

Let $\lambda \in \mathbb{Q}$ (or a suitable scalar depending on the specific structure of Q) such that $\lambda \neq 0, 1$. We can construct a map that "scales" the generators of G modulo the center:

- On the quotient Q , the map induces $\bar{x} \mapsto \lambda\bar{x}$ (using the additive notation for the abelian group Q).
- To preserve the commutator relation $[x, y]$, the center $Z(G)$ must be scaled by λ^2 .

Such a map, call it ψ_λ , is an automorphism of G . Crucially, consider its action on the quotient Q :

$$\psi_\lambda(\bar{x}) = \lambda\bar{x} \neq \bar{x} \quad (\text{since } \lambda \neq 1)$$

Because ψ_λ does not induce the identity map on $G/Z(G)$, it is **not** a central automorphism.

- Assumption that $\text{Aut}(G)$ is abelian $\implies$ All automorphisms are central ($\text{Aut}(G) = \text{Aut}_c(G)$).
- Structure of torsion-free non-abelian $G \implies$ Existence of non-central (scaling) automorphisms.

This is a contradiction. Thus, G cannot be non-abelian.

1.4 Conclusion

We have exhausted all possibilities. If G is abelian, the torsion-free property of A forces G to be an elementary abelian 2-group, making $\text{Aut}(G)$ non-abelian (or trivial). If G is non-abelian, the abelian property of A forces rigidity ($\text{Aut}(G) = \text{Aut}_c(G)$) that is incompatible with the flexible scaling automorphisms allowed by the torsion-free structure.

Therefore, no group G exists with $\text{Aut}(G) \cong \mathbb{Q}$, $\text{Aut}(G) \cong \mathbb{R}$, or any other non-trivial torsion-free abelian group.

Exercise 1.4.1

1. Show that if G is a group of exponent 2, then G is abelian.
2. Prove that for the Heisenberg group $H(\mathbb{Z})$, $\text{Aut}(H(\mathbb{Z}))$ is not abelian by exhibiting a non-central automorphism.
3. Verify that the inversion map $x \mapsto x^{-1}$ is an automorphism if and only if the group is abelian.

Clarifications on the Proof

1. The Ordering: $\alpha(g)g^{-1}$ vs $g^{-1}\alpha(g)$

You are correct to spot this. In the derivation of the contradiction, we derived that $\alpha(g)$ and g must commute modulo the center. Technically, from $\alpha(g)y\alpha(g)^{-1} = gyg^{-1}$, we deduce that the element $z = g^{-1}\alpha(g)$ commutes with every $y \in G$, placing z in the center $Z(G)$.

Why are $\alpha(g)g^{-1}$ and $g^{-1}\alpha(g)$ interchangeable here?

1. Note that $\alpha(g)g^{-1} = g(g^{-1}\alpha(g))g^{-1}$.
2. This shows that $\alpha(g)g^{-1}$ is the *conjugate* of $g^{-1}\alpha(g)$ by g .
3. Since we established that $g^{-1}\alpha(g)$ is in the Center $Z(G)$, and central elements are invariant under conjugation, it follows that:

$$\alpha(g)g^{-1} = g^{-1}\alpha(g)$$

So, while $g^{-1}\alpha(g)$ is the standard definition for the cohomology derivation, they represent the same element in this specific context.

2. Why is τ_α a 1-cocycle?

We claimed that the map $\tau_\alpha(x) = x^{-1}\alpha(x)$ is a derivation (or 1-cocycle) from G to $Z(G)$. A map $f : G \rightarrow M$ (where G acts on the module M) is a 1-cocycle if it satisfies the condition:

$$f(xy) = f(x) + x \cdot f(y)$$

(using additive notation for the module M).

In our case, $M = Z(G)$. The action of G on its own center is **trivial** (i.e., $g \cdot z = z$ for all $g \in G, z \in Z(G)$). Thus, the condition simplifies to the standard homomorphism property (written multiplicatively):

$$\tau(xy) = \tau(x)\tau(y)$$

Let's verify this for $\tau_\alpha(x) = x^{-1}\alpha(x)$:

$$\begin{aligned} \tau_\alpha(xy) &= (xy)^{-1}\alpha(xy) \\ &= y^{-1}x^{-1}\alpha(x)\alpha(y) \end{aligned}$$

We know that $x^{-1}\alpha(x)$ is in the center $Z(G)$. Central elements commute with everything, so we can swap $x^{-1}\alpha(x)$ with y^{-1} :

$$\begin{aligned} &= y^{-1}[x^{-1}\alpha(x)]\alpha(y) \\ &= [x^{-1}\alpha(x)][y^{-1}\alpha(y)] \\ &= \tau_\alpha(x)\tau_\alpha(y) \end{aligned}$$

Thus, τ_α is a homomorphism from G to $Z(G)$, which makes it a 1-cocycle for the trivial action.

3. Rank of Q

When we speak of the **rank** of the torsion-free abelian group $Q = G/Z(G)$, we refer to the **Torsion-Free Rank** (also known as the $\mathbb{Q}$ -rank). Ideally, if we view Q as an additive group, the rank is the dimension of the vector space $Q \otimes_{\mathbb{Z}} \mathbb{Q}$.

$$\text{rank}(Q) = \dim_{\mathbb{Q}}(Q \otimes \mathbb{Q})$$

Intuitively, this is the size of the largest subset of elements in Q that are linearly independent over the integers (i.e., no linear combination with integer coefficients equals 0).

Why Rank ≥ 2 ? If the rank of Q is 1, then Q is isomorphic to a subgroup of $\mathbb{Q}$. As shown in the simpler case, subgroups of $\mathbb{Q}$ are **locally cyclic** (every finitely generated subgroup is cyclic). If $G/Z(G)$ is locally cyclic, then G must be abelian.

- *Sketch:* For any $x, y \in G$, their projections $\bar{x}, \bar{y}$ in a locally cyclic quotient generate a cyclic subgroup. Thus $\bar{x} = \bar{g}^n, \bar{y} = \bar{g}^m$. This implies x and y are powers of the same element g modulo the center, so they commute.

Since we are in the **Non-Abelian Case**, Q cannot be rank 1. Thus, $\text{rank}(Q) \geq 2$.

4. Why is the Commutator Non-Trivial?

The commutator map $\omega : Q \times Q \rightarrow Z(G)$ is defined by $\omega(\bar{x}, \bar{y}) = xyx^{-1}y^{-1}$. If this map were trivial (i.e., the image is always the identity e), then:

$$xyx^{-1}y^{-1} = e \implies xy = yx \quad \forall x, y \in G$$

This would imply G is abelian. Since we assume G is ****non-abelian****, there must exist at least one pair x, y that does not commute, making the commutator map non-trivial.

5. The Existence of ψ_λ

This is the most subtle point. We cannot define x^λ (for $\lambda \in \mathbb{Q}$) in just *any* group. However, the assumption that $\text{Aut}(G)$ is a large torsion-free divisible group (like $\mathbb{Q}$ or $\mathbb{R}$) implies strong structural properties on G .

Specifically, G must be a subgroup of a ****divisible nilpotent group**** (specifically, its *Mal'cev Completion*). Divisible nilpotent groups behave exactly like Lie algebras over $\mathbb{Q}$. In such groups, we can define rational powers g^λ .

Let's look at a concrete example using the simplest non-abelian group of this type: the ****Heisenberg Group**** H .

$$H = \langle x, y, z \mid [x, y] = z, [x, z] = [y, z] = 1 \rangle$$

Here, $Z(G) = \langle z \rangle$ and $Q = G/Z(G) \cong \mathbb{Z} \times \mathbb{Z}$ (generated by $\bar{x}, \bar{y}$). We want to define an automorphism that scales x but not y . Let $\lambda = 2$. We cannot simply say $\psi(x) = x^2$ and $\psi(y) = y$ because that might break the commutator relation. Let's verify the commutator relation $[x^2, y]$. Using the identity $[a^n, b] = [a, b]^n$ (which holds when $[a, b]$ is central):

$$[x^2, y] = [x, y]^2 = z^2$$

So, if we define the map as:

$$\psi(x) = x^2, \quad \psi(y) = y, \quad \psi(z) = z^2$$

This preserves the defining relation $\psi([x, y]) = \psi(z) = z^2$ and $[\psi(x), \psi(y)] = [x^2, y] = z^2$. Thus, ψ is a valid endomorphism. In the context of divisible groups (where we can take roots), we can inverse this with $\lambda = 1/2$, so it is an automorphism.

The Contradiction: Is ψ a central automorphism? Check its action on the quotient Q :

$$\psi(\bar{x}) = \overline{x^2} = 2\bar{x} \neq \bar{x}$$

Since ψ acts non-trivially on the quotient (it doubles the generator $\bar{x}$), it is ****not**** a central automorphism. This contradicts the earlier deduction that $\text{Aut}(G) \text{ abelian} \implies \text{Aut}(G) = \text{Aut}_c(G)$.

2

lie algebra stuff

2.1 Step 2: The Lie Algebra Connection

To find a contradiction, we must show that there exists an automorphism that is *not* central. To do this, we need to understand the structure of G . Since $\text{Aut}(G) \cong \mathbb{Q}$ (or $\mathbb{R}$), G must be a divisible group (we can take n -th roots of elements). This invokes a powerful correspondence in algebra.

Theorem 2.1.1: Mal'cev Correspondence

There is a bijective correspondence between divisible torsion-free nilpotent groups and rational nilpotent Lie algebras.

For our purposes, this means we can study G by studying a corresponding Lie Algebra $\mathfrak{g}$ over the field $\mathbb{Q}$.

- The Lie Algebra $\mathfrak{g}$ is a vector space over $\mathbb{Q}$ equipped with a bracket $[X, Y]$.
- There is a bijective map $\exp : \mathfrak{g} \rightarrow G$. Unlike in rotation groups (where $\exp$ wraps around), for nilpotent groups this map is a **global bijection** (diffeomorphism).
- Every element $g \in G$ has a unique logarithm $X \in \mathfrak{g}$ such that $g = \exp(X)$.

The Campbell-Baker-Hausdorff (CBH) Formula

You asked how the multiplication in G is determined by the algebra. The answer is the **CBH Formula**:

$$\exp(X) \cdot \exp(Y) = \exp \left(X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] - \frac{1}{12}[Y, [X, Y]] + \dots \right)$$

Since $\text{Aut}(G)$ is abelian, G is nilpotent of **class 2**. This means all commutators of commutators vanish ($[X, [Y, Z]] = 0$). Consequently, the infinite series truncates at the second term:

$$\exp(X) \cdot \exp(Y) = \exp\left(X + Y + \frac{1}{2}[X, Y]\right)$$

This formula tells us that group multiplication is just vector addition plus a "twist" given by the commutator.

2.1.1 Step 3: Constructing the Scaling Automorphism

We can now define coordinates. Since $\mathfrak{g}$ is a vector space, let $\{X, Y\}$ be basis vectors for the quotient $\mathfrak{g}/Z(\mathfrak{g})$ and $Z = [X, Y]$ be the basis for the center. Any $g \in G$ corresponds to a unique vector $v = aX + bY + cZ$. We can represent g as (a, b, c) .

We define a scaling map ψ_λ for some scalar $\lambda \in \mathbb{Q}$ ($\lambda \neq 0, 1$). In the Lie Algebra (vector space), scaling is easy. We scale the generators X and Y by λ .

$$X \mapsto \lambda X, \quad Y \mapsto \lambda Y$$

What happens to the center vector Z ? Since $Z = [X, Y]$ and the bracket is bilinear:

$$[\lambda X, \lambda Y] = \lambda^2[X, Y] = \lambda^2 Z$$

So we must scale the center component by λ^2 .

Let us lift this map to the group G . Define $\psi_\lambda : G \rightarrow G$ by applying this scaling to the coordinates:

$$\psi_\lambda(\exp(aX + bY + cZ)) = \exp(\lambda aX + \lambda bY + \lambda^2 cZ)$$

Is this a group homomorphism? We verify using the truncated CBH formula. Let $g_1 \leftrightarrow A$ and $g_2 \leftrightarrow B$.

$$\psi_\lambda(g_1 \cdot g_2) = \psi_\lambda(\exp(A + B + \frac{1}{2}[A, B]))$$

Since ψ_λ is linear on vector components and quadratic on the bracket term:

$$= \exp\left(\lambda(A + B) + \lambda^2\left(\frac{1}{2}[A, B]\right)\right)$$

(Note: The bracket $[A, B]$ lies in the center, so it scales by λ^2).

Now compute $\psi_\lambda(g_1) \cdot \psi_\lambda(g_2)$. Let $A' = \lambda A$ (on generators) and $B' = \lambda B$.

$$\psi_\lambda(g_1)\psi_\lambda(g_2) = \exp(A') \cdot \exp(B') = \exp\left(A' + B' + \frac{1}{2}[A', B']\right)$$

Substituting the scaling:

$$= \exp\left(\lambda A + \lambda B + \frac{1}{2}[\lambda A, \lambda B]\right) = \exp\left(\lambda(A + B) + \frac{1}{2}\lambda^2[A, B]\right)$$

The results match. Thus, ψ_λ is a homomorphism. Since $\lambda \neq 0$ and G is divisible, it is invertible, hence an automorphism.

2.1.2 Step 4: The Contradiction

We check if ψ_λ is a central automorphism. Consider the generator $g = \exp(X)$.

$$\psi_\lambda(g) = \exp(\lambda X) = g^\lambda$$

In the quotient group $G/Z(G)$, this maps the coset $\bar{g}$ to $\bar{g}^\lambda$. If $\lambda \neq 1$, then $\bar{g}^\lambda \neq \bar{g}$. Therefore, ψ_λ acts non-trivially on the quotient. It is **not** a central automorphism.

Conclusion:

- Assumption ($\text{Aut}(G)$ abelian) $\implies$ All automorphisms are central.
- Structure (Divisible Nilpotent) $\implies$ There exist non-central scaling automorphisms.

This is a contradiction. Thus, no such group G exists.

3

another idea

We prove that there does not exist a group G such that $\text{Aut}(G)$ is isomorphic to a subgroup of $\mathbb{Q}^n$ (or any non-trivial torsion-free divisible abelian group). The proof relies on analyzing the structure of the inner automorphism group and constructing a specific "inversion" automorphism that leads to a contradiction.

3.1 Introduction

Let A be a non-trivial torsion-free divisible abelian group (e.g., $A = \mathbb{Q}$, $A = \mathbb{R}$, or $A = \mathbb{Q}^n$). We aim to prove that there is no group G such that $\text{Aut}(G) \cong A$.

3.2 Proof

Theorem 3.2.1: main theorem

There is no group G such that $\text{Aut}(G) \cong A$, where A is a subgroup of $\mathbb{Q}^n$ containing at least one non-zero element.

Proof. Suppose such a group G exists. Let $\mathcal{A} = \text{Aut}(G) \cong A$.

Step 1: Structural Consequences for G

Since $\mathcal{A}$ is abelian, every subgroup of $\mathcal{A}$ is normal. In particular, the group of inner automorphisms, $\text{Inn}(G) \cong G/Z(G)$, is a normal subgroup of $\mathcal{A}$.

1. Since $\mathcal{A}$ is abelian, $\text{Inn}(G)$ is abelian. Thus, $G/Z(G)$ is abelian, which implies that G is nilpotent of class at most 2. The commutator subgroup G' is contained in the center $Z(G)$.
2. Since $\mathcal{A}$ is torsion-free (being a subgroup of $\mathbb{Q}^n$), $\text{Inn}(G)$ is torsion-free. Thus, the quotient $Q = G/Z(G)$ is a torsion-free abelian group.

Step 2: Constraint on Automorphisms

Since $\mathcal{A}$ is abelian, every automorphism $\alpha \in \mathcal{A}$ must commute with every inner automorphism $c_g \in \text{Inn}(G)$ (where $c_g(x) = gxg^{-1}$).

$$\alpha \circ c_g = c_g \circ \alpha$$

Applying this to an arbitrary element $x \in G$:

$$\begin{aligned}\alpha(gxg^{-1}) &= g\alpha(x)g^{-1} \\ \alpha(g)\alpha(x)\alpha(g)^{-1} &= g\alpha(x)g^{-1}\end{aligned}$$

Since α is surjective, let $y = \alpha(x)$. As x ranges over G , y ranges over G .

$$\alpha(g)y\alpha(g)^{-1} = gyg^{-1} \quad \forall y \in G$$

Multiplying by g^{-1} on the left and $\alpha(g)$ on the right:

$$(g^{-1}\alpha(g))y = y(g^{-1}\alpha(g))$$

This implies that the element $g^{-1}\alpha(g)$ commutes with every element $y \in G$. Therefore, $g^{-1}\alpha(g) \in Z(G)$, or equivalently:

$$\alpha(g) \equiv g \pmod{Z(G)} \quad \forall g \in G$$

Thus, every automorphism of G is a **central automorphism**; it induces the identity map on the quotient group $G/Z(G)$.

Step 3: Construction of the Inversion Automorphism Ψ

We define a specific map $\Psi : G \rightarrow G$ and show it is an automorphism that violates the constraint established in Step 2.

Let $Q = G/Z(G)$ and $Z = Z(G)$. Since G is a central extension of Z by Q , elements of G can be identified with pairs (a, z) where $a \in Q$ and $z \in Z$, with the multiplication rule:

$$(a, z_1) \cdot (b, z_2) = (a + b, z_1 + z_2 + f(a, b))$$

where $f : Q \times Q \rightarrow Z$ is a 2-cocycle (factor set). (We use additive notation for the abelian groups Q and Z).

According to the theory of nilpotent groups (see Warfield, *Nilpotent Groups*, Theorem 5.4), for torsion-free groups, the factor set f corresponds to an alternating bilinear form (the commutator map). Bilinearity implies:

$$f(-a, -b) = (-1)(-1)f(a, b) = f(a, b)$$

We define the map $\Psi : G \rightarrow G$ by:

$$\Psi((a, z)) = (-a, z)$$

This map inverts elements in the quotient Q and fixes elements in the center Z .

Verification that Ψ is a Homomorphism:

$$\begin{aligned}\Psi((a, z_1) \cdot (b, z_2)) &= \Psi(a + b, z_1 + z_2 + f(a, b)) \\ &= (-(a + b), z_1 + z_2 + f(a, b))\end{aligned}$$

$$\begin{aligned}\Psi(a, z_1) \cdot \Psi(b, z_2) &= (-a, z_1) \cdot (-b, z_2) \\ &= (-a - b, z_1 + z_2 + f(-a, -b))\end{aligned}$$

Since $f(a, b) = f(-a, -b)$, the map preserves the group operation. Since $\Psi^2 = \text{id}$, it is bijective and thus an automorphism.

Step 4: The Contradiction

We have established two contradictory facts about Ψ :

1. **From Step 2:** Since $\Psi \in \text{Aut}(G)$ and $\text{Aut}(G)$ is abelian, Ψ must be a central automorphism. It must induce the identity map on Q .

$$\Psi(a, z) \equiv (a, z) \pmod{Z} \implies -a = a$$

2. **From Step 3:** By definition, Ψ induces the inversion map on Q .

$$\Psi(a, z) = (-a, z)$$

These conditions require $-a = a$, or $2a = 0$, for all $a \in Q$. Since Q is a subgroup of $\mathbb{Q}^n$, it is **torsion-free**. The only solution to $2a = 0$ is $a = 0$. This implies Q is the trivial group, so $G = Z(G)$, and G is abelian.

The Abelian Case: If G is abelian, the map $\varphi : x \mapsto x^{-1}$ is an automorphism.

$$\varphi(x^{-1}) = (x^{-1})^{-1} = x \implies \varphi^2 = \text{id}$$

Thus $\text{Aut}(G)$ contains an element of order 2. But $\text{Aut}(G) \cong A \leq \mathbb{Q}^n$, which is torsion-free. Therefore $\varphi = \text{id}$, which implies $x^{-1} = x$ for all x . G must be an elementary abelian 2-group. The automorphism group of an elementary abelian 2-group of rank $k > 1$ is $\text{GL}(k, \mathbb{F}_2)$, which is non-abelian. If rank $k = 1$, $\text{Aut}(G)$ is trivial. Since A is non-trivial and abelian, this is a contradiction.

□

4

another attempt

4.1 Theorem

There does not exist a group G such that $\text{Aut}(G) \cong A$, where A is a subgroup of $\mathbb{Q}^n$ (or any non-trivial torsion-free divisible abelian group).

4.2 Proof

Assume for the sake of contradiction that such a group G exists. Let $\mathcal{A} = \text{Aut}(G)$. By hypothesis, $\mathcal{A}$ is torsion-free and abelian.

Step 1: Structural Reduction

Since $\mathcal{A}$ is abelian, the inner automorphism group $\text{Inn}(G) \cong G/Z(G)$ is a subgroup of $\mathcal{A}$.

1. $G/Z(G)$ is abelian, implying G is nilpotent of class at most 2.
2. $G/Z(G)$ is a subgroup of $\mathbb{Q}^n$, so it is a ****torsion-free**** abelian group.

Let $Q = G/Z(G)$ and $Z = Z(G)$. The group G is a central extension of Z by Q .

Step 2: The Centrality Constraint

Since $\mathcal{A}$ is abelian, every automorphism $\alpha \in \mathcal{A}$ must commute with every inner automorphism c_g .

$$\alpha \circ c_g = c_g \circ \alpha \implies \alpha(gxg^{-1}) = g\alpha(x)g^{-1}$$

This implies $\alpha(g)y\alpha(g)^{-1} = gyg^{-1}$ for all y , forcing $g^{-1}\alpha(g) \in Z$. Thus, every automorphism is **central**:

$$\alpha(g) \equiv g \pmod{Z} \quad \forall \alpha \in \mathcal{A}, g \in G$$

In particular, every automorphism must induce the **Identity Map** on the quotient Q .

Step 3: The Factor Set and Commutator

We construct a specific automorphism to contradict Step 2. By Warfield (*Nilpotent Groups*, Theorem 5.4), the extension class of G in $H^2(Q, Z)$ maps to a commutator form $\omega \in \text{Hom}(\Lambda^2 Q, Z)$.

$$\omega(u, v) = [u, v]$$

The theorem states that ω is an ****Alternating Bilinear Form****. Bilinearity implies:

$$\omega(-u, -v) = (-1)(-1)\omega(u, v) = \omega(u, v)$$

This signifies that the commutator relations are invariant if we invert the generators in Q .

Step 4: The Inversion Automorphism

We define a map $\Psi : G \rightarrow G$. Representing elements of G as pairs (q, z) with factor set f :

$$\Psi((q, z)) = (-q, z)$$

(Invert the quotient, fix the center). We verify Ψ is a homomorphism. The condition is $f(u, v) = f(-u, -v)$.

1. The antisymmetric part of f is the commutator ω , which satisfies $\omega(u, v) = \omega(-u, -v)$ by bilinearity (Step 3).
2. The symmetric part of f corresponds to an abelian extension. Since Q is torsion-free, we can choose a symmetric bilinear cocycle, or observe that for abelian groups, inversion is always an automorphism.

Thus, Ψ is a well-defined automorphism of G .

Step 5: The Contradiction

We now have an automorphism $\Psi \in \text{Aut}(G)$.

1. **Torsion:** $\Psi^2(q, z) = (-(-q), z) = (q, z)$. So $\Psi^2 = \text{id}$.
2. **Non-Centrality:** By definition, Ψ acts as -1 on the quotient Q . From Step 2, all automorphisms must act as $+1$ on Q . This forces $-1 = +1$ on Q .

$$-q = q \implies 2q = 0$$

Since Q is torsion-free, $q = 0$.

3. **Collapse:** This implies Q is trivial, so $G = Z$, and G is Abelian.
4. **Abelian Case:** If G is Abelian, the map $\varphi(g) = g^{-1}$ is an automorphism of order 2. Since $\text{Aut}(G) \leq \mathbb{Q}^n$ is torsion-free, φ must be identity. $g^{-1} = g \implies g^2 = 1$. G is an elementary abelian 2-group. $\text{Aut}(G) \cong \text{GL}(k, \mathbb{F}_2)$. For $k \geq 2$, this is non-abelian. For $k = 1$, it is trivial. In neither case is it isomorphic to a non-trivial subgroup of $\mathbb{Q}^n$ (which is abelian and infinite).

Conclusion: No such group exists.

Bibliography

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